

gdTAI V4.2 NK definition repair

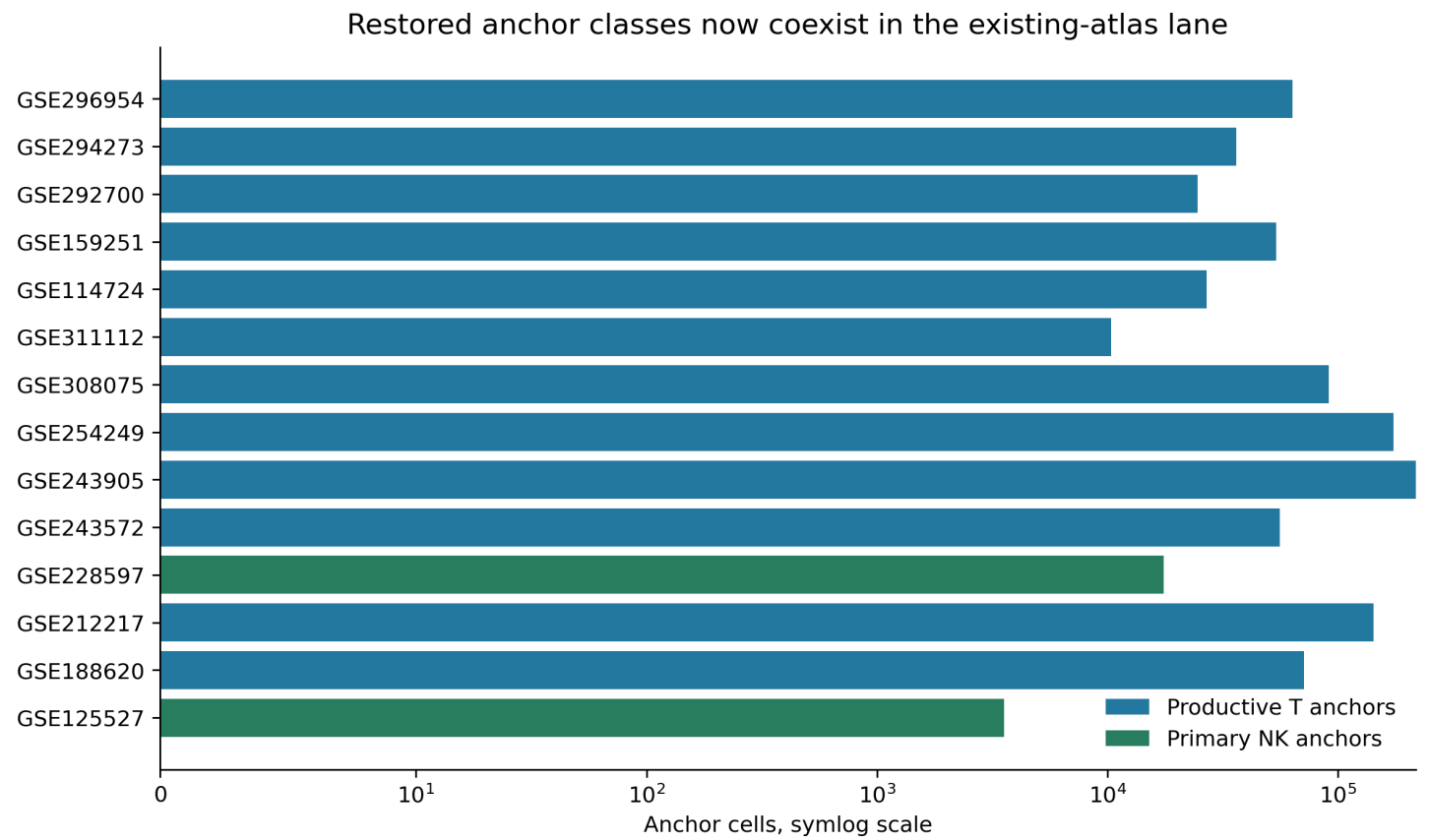
This read-only repair addresses the cohort-confounded pseudo-NK failure. It does not alter the canonical atlas, refit scVI, train gdTAI, or use locked cohorts for threshold selection.

Result: PASS_NK_REFERENCE_READY. The repaired definition identifies 469 cells with both latent and strict lineage evidence. The source-balanced subset is ready for low-weight development use with capped source influence.

762,280	21,054	469	469
restored existing-atlas T anchors	dual-annotation NK anchors	two-evidence pseudo-NK cells	capped-weight training rows

1. Defect repaired

The prior sidecar mapped no productive-T anchors into the existing-atlas lane even though its source object contains 762,280 cells with productive TRA or TRB evidence. Restoring these cells means NK and T anchors now coexist within the same old-atlas lane; productive-T anchors also remain present in every new cohort.



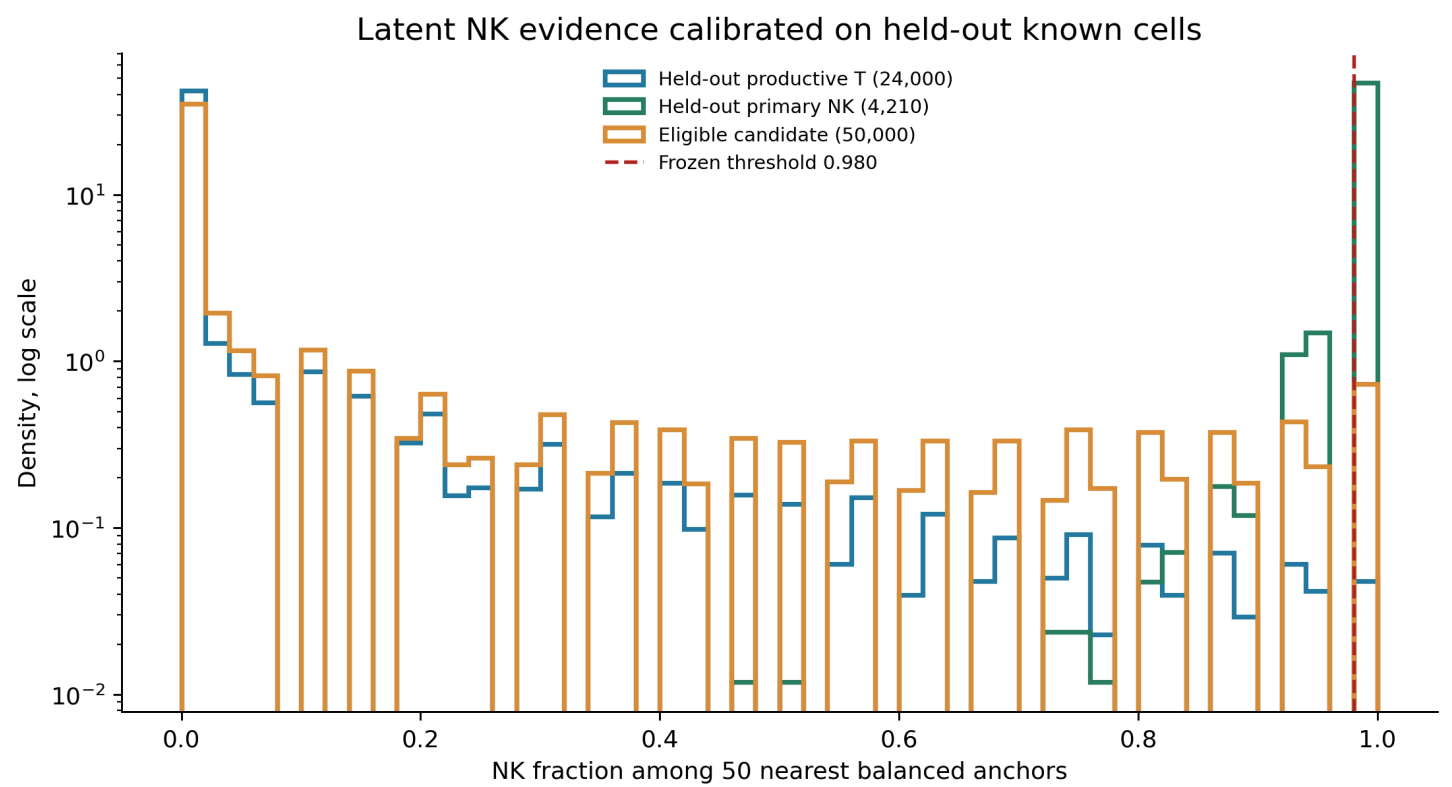
Anchor counts use expression-independent TCR metadata and the frozen dual-annotation NK labels.

source_gse_id	n_primary_nk	n_productive_t	input_lane
GSE114724	0	26893	new_cohort
GSE125527	3547	0	existing_atlas
GSE159251	0	53773	new_cohort
GSE188620	0	71160	existing_atlas
GSE212217	0	142604	existing_atlas

source_gse_id	n_primary_nk	n_productive_t	input_lane
GSE228597	17507	0	existing_atlas
GSE243572	0	55770	existing_atlas
GSE243905	0	217164	existing_atlas
GSE254249	0	174010	existing_atlas
GSE292700	0	24577	new_cohort
GSE294273	0	36221	new_cohort
GSE296954	0	63405	new_cohort
GSE308075	0	91239	existing_atlas
GSE311112	0	10333	existing_atlas

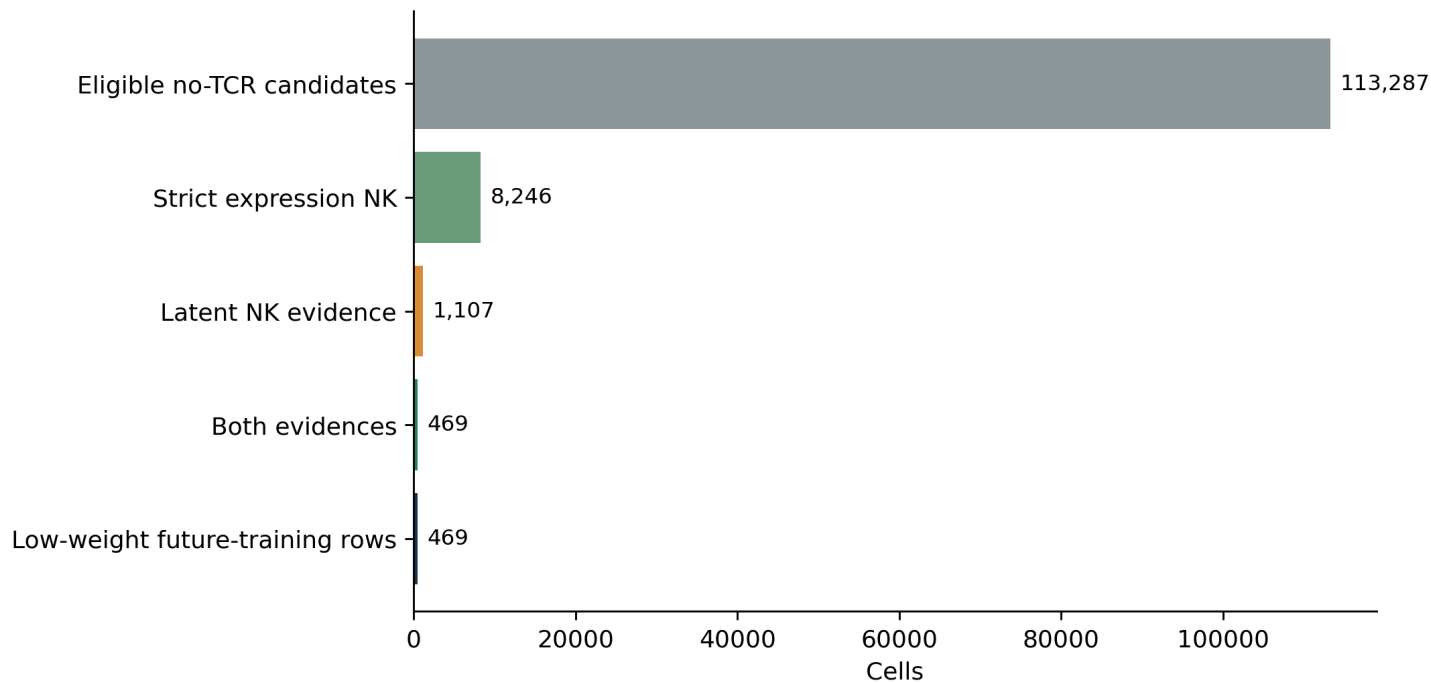
2. NK definition

A development cell is called **NK_CONFIDENT** only when it has no productive TCR, no doublet flag, passes the latent nearest-anchor threshold, remains within the known-NK distance envelope, expresses both `FCER1G` and `TYROBP` plus at least one of `KLRD1/NCR1/FCGR3A/KLRC1` , and has no detected `CD3D/CD3E/CD3G/TRAT1/BCL11B` . Cytotoxic genes are audit-only and cannot establish NK identity.



The threshold is the lowest discrete 50-neighbor score whose empirical held-out productive-T FPR is at most 0.1%; score ties fail closed.

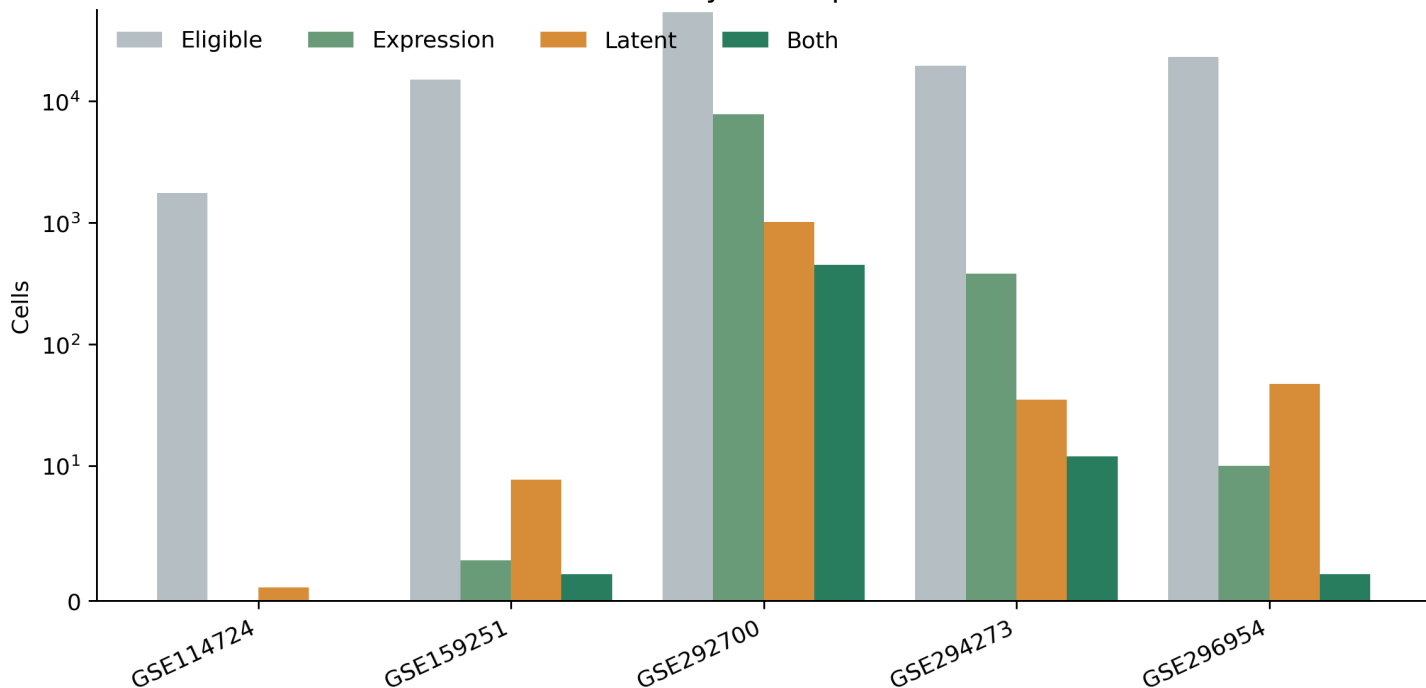
Conservative NK evidence funnel



Only the intersection of independent latent and lineage-expression evidence is accepted.

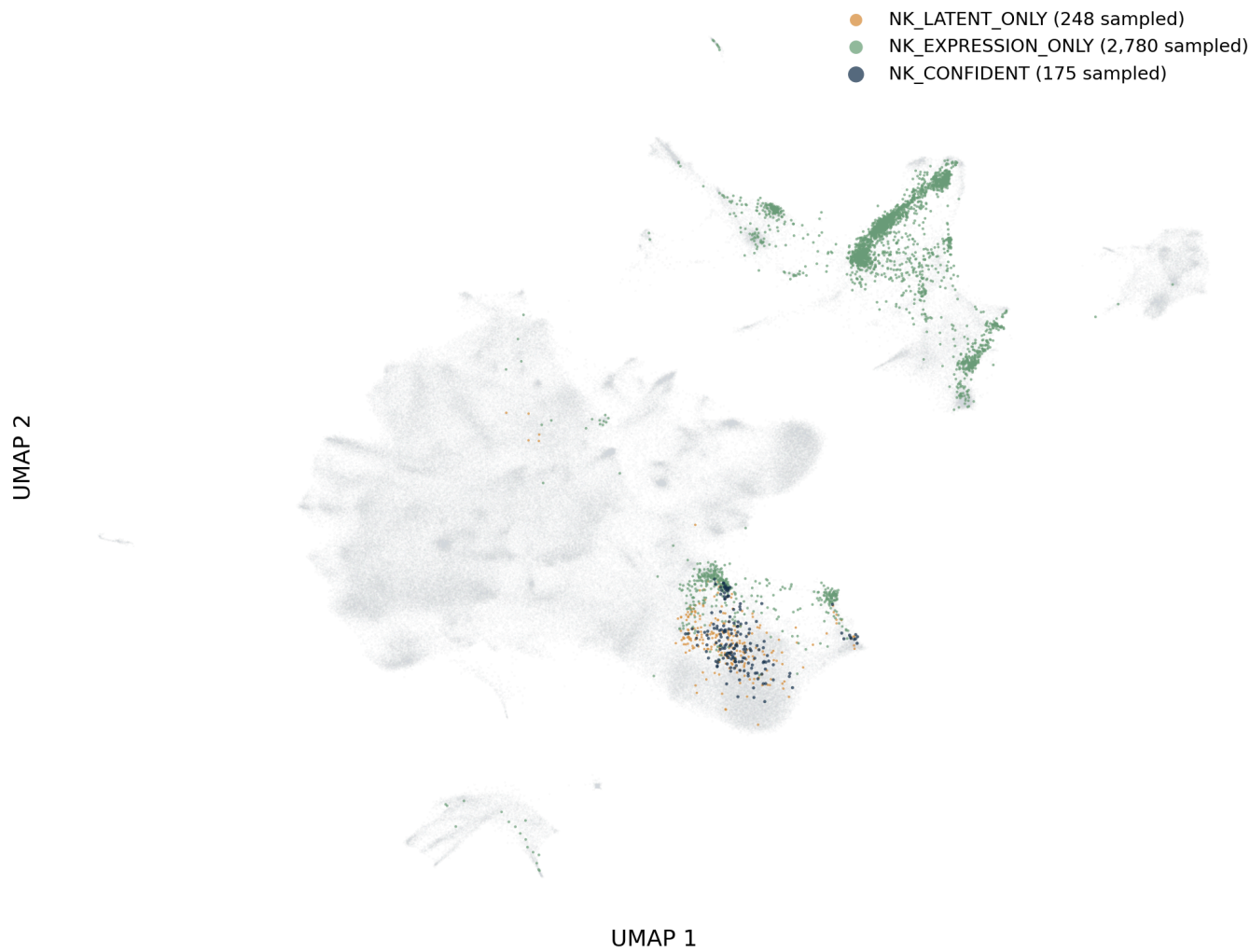
3. Source-level result

NK evidence by development source



All development sources are reported separately; source imbalance is controlled by capped effective weights for any future low-weight training use.

Repaired NK evidence on the existing diagnostic UMAP



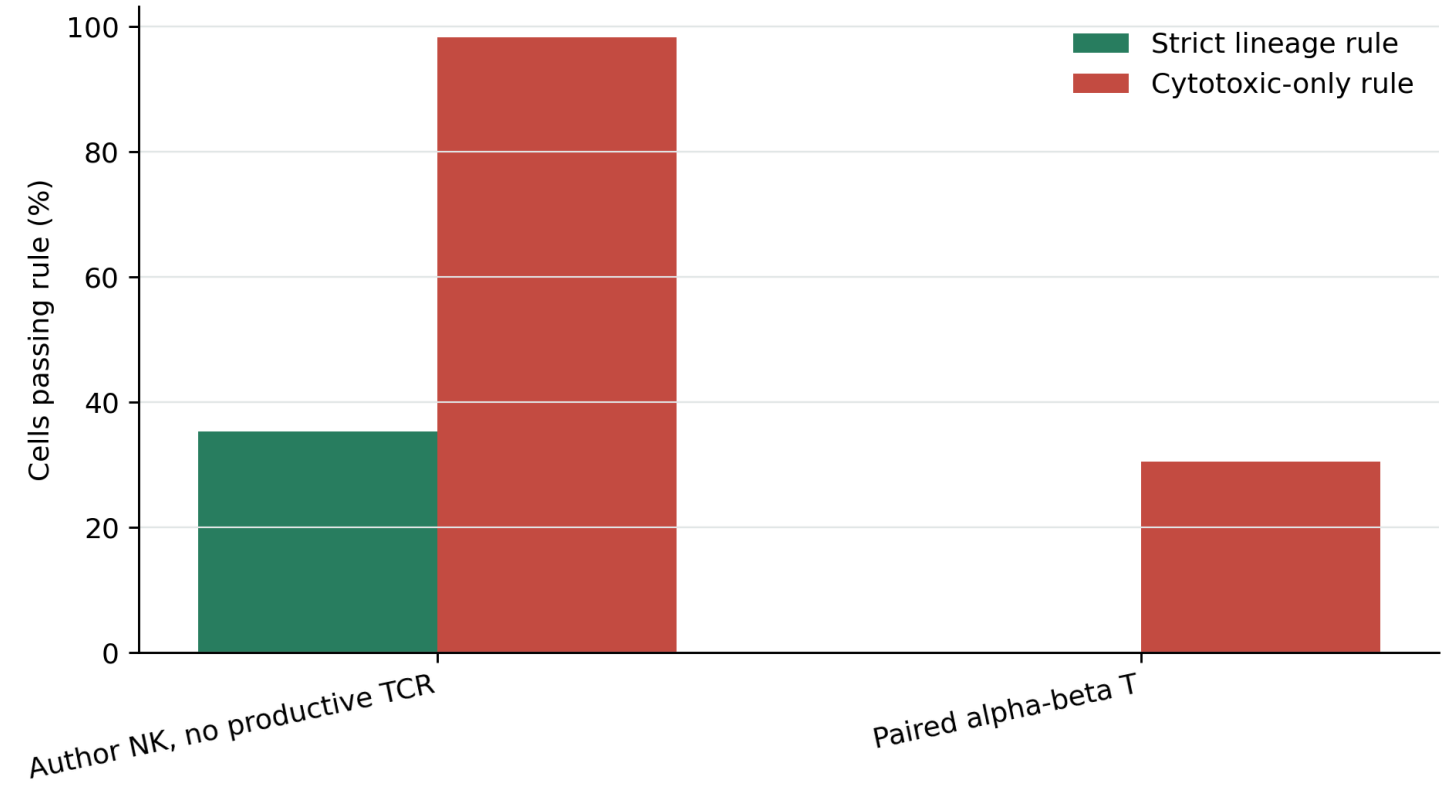
Diagnostic UMAP overlay; the full label table, not the visualization sample, defines counts.

source_gs_e_id	n_candidates	n_expression	n_latent	n_confident	n_training	effective_training_weight	n_donors	n_libraries	expression_rate	latent_rate	confident_rate	training_fraction	effective_training_source_fraction
GSE114724	1759	0	1	0	0	0.000000	3	5	0.00%	0.06%	0.00%	0.00%	0.00%
GSE159251	15125	3	9	2	2	0.500000	4	9	0.02%	0.06%	0.01%	0.01%	3.75%
GSE292700	53677	7850	1015	453	453	9.333333	2	6	14.62%	1.89%	0.84%	0.84%	70.00%
GSE294273	19523	383	35	12	12	3.000000	12	12	1.96%	0.18%	0.06%	0.06%	22.50%
GSE296954	23203	10	47	2	2	0.500000	11	22	0.04%	0.20%	0.01%	0.01%	3.75%

4. Locked validation and specificity

GSE169246 was not integrated and did not control any threshold. Its author-defined NK cells and paired alpha-beta T cells test only the fixed strict expression rule. The cytotoxic-only comparator demonstrates why NKG7/GNLY-like activity is not accepted as NK identity.

Locked GSE169246: lineage evidence versus cytotoxicity



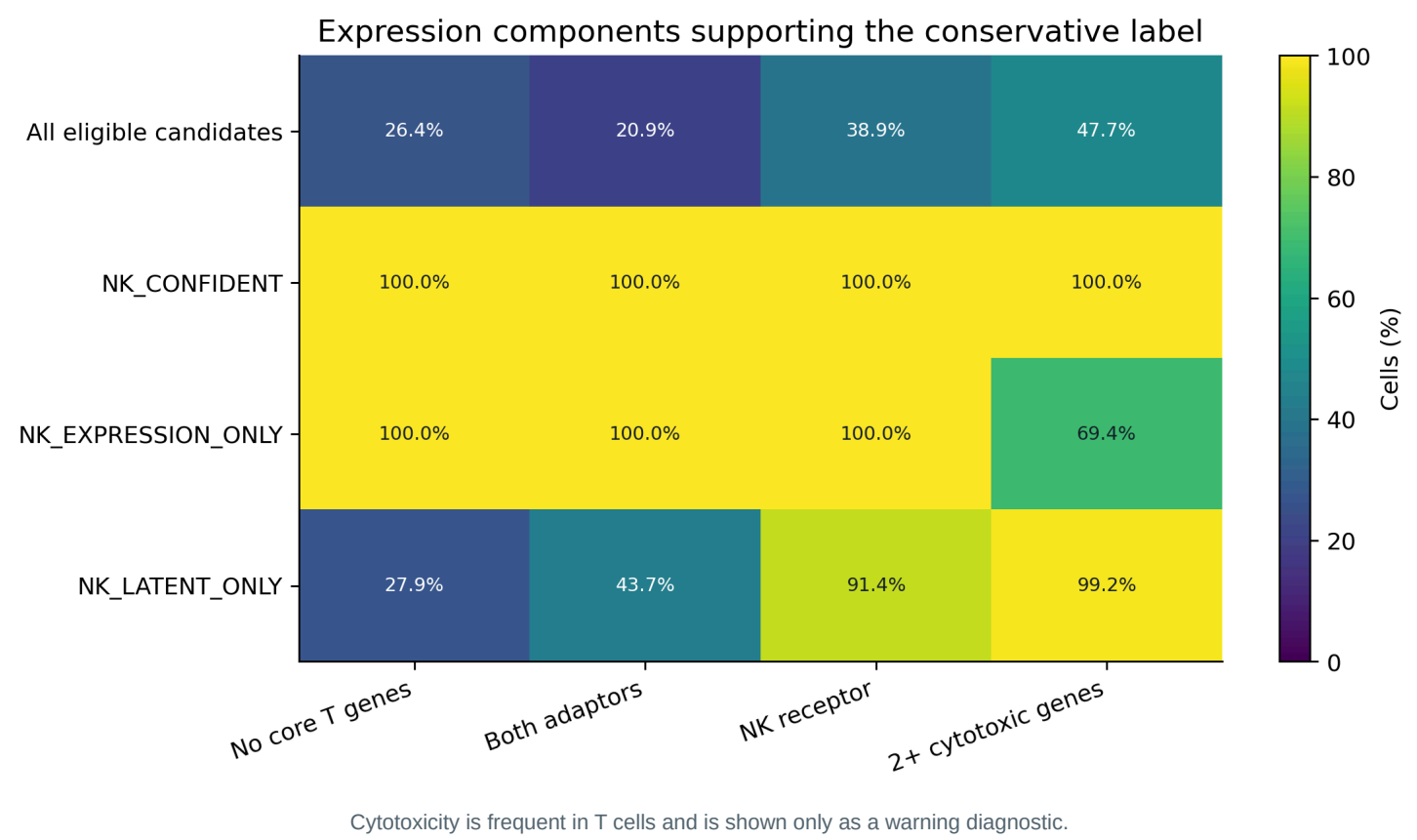
Fixed-rule passage in locked author-NK and paired-abT populations.

group	n_cells	n_strict_expression_nk	strict_expression_nk_rate	n_cytotoxic_only	cytotoxic_only_rate
Author NK, no productive TCR	7770	2744	35.32%	7631	98.21%
Paired alpha-beta T	54925	69	0.13%	16699	30.40%

Development productive-T controls

cohort_id	n_productive_t	strict_expression_nk_rate	cytotoxic_only_rate
GSE114724	26893	0.00%	33.08%
GSE159251	53773	0.00%	37.38%
GSE292700	24577	0.46%	40.48%
GSE294273_GSE294274	36221	0.02%	30.49%
GSE296954	63405	0.00%	54.25%

5. Marker components



Scope: these are conservative NK reference labels, not a finished gdTAI V4 model. No classifier was fitted, no threshold was tuned on GSE169246 or the candidate counts, no H5AD was changed, and nothing was pushed to GitHub.

Reproducibility

- Latent SHA-256: 7030b28be885b85a4b24ce2223bb16f7c7288a29195f2512bf7bbba330791ccf
- Script SHA-256: c282b0d106caa3561b8f9c526c5dccfea522d610aad64e3ad99e53ea8863a262
- Random seed: 20260817
- Neighbor search: exact_gpu_brute
- Bit-identical exact-GPU repeat: true
- GPU: NVIDIA A100 80GB PCIe ; CPU fallback: no